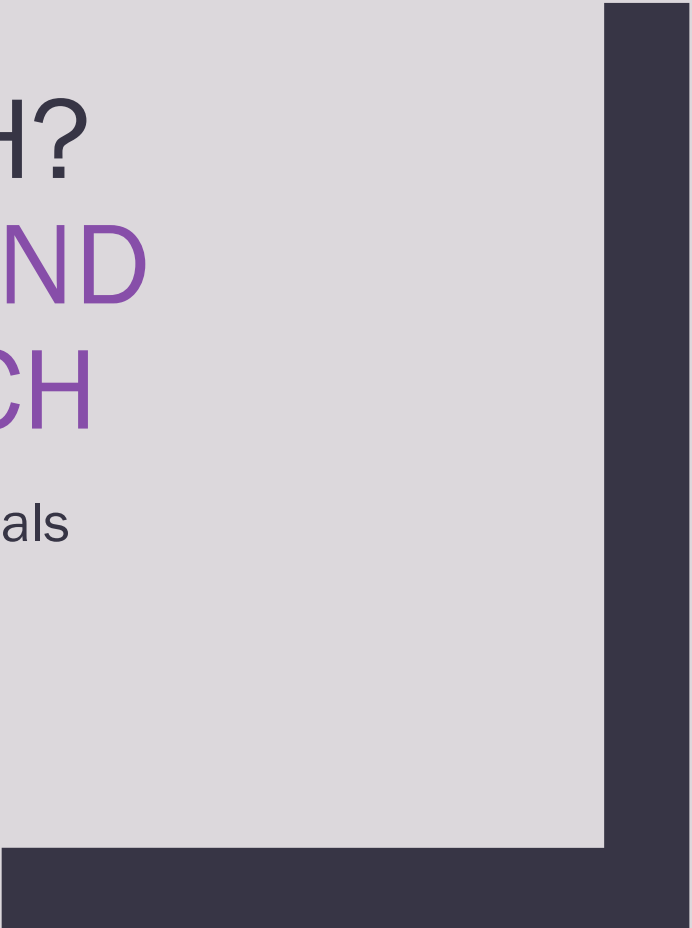




WHAT IS RESEARCH? CHARACTERISTICS AND TYPES OF RESEARCH

Research Methodology for IT Professionals

Lecture 3, 17 February 2020

Lynn Coleman

Activity One – own your definition

- How would you define or what do you understand the process of academic/scientific research to be?
 - I would define research as...
 - My understanding of research is that it is...
1. Use notes, reading, lecture slides
 2. Share definition with person next you
 3. After this short 'consultation' re-write your statement if your understanding/definition was not accurate or unclear

Housekeeping

- BLACKBOARD – updates? Any problems
- Register – as per normal
- How to best use time during classroom sessions
- **Consultations**
 - *3:30 – 4:15 on Monday or by appointment*



Ranji Kumar – Chapter 1

Research: A way of thinking

- What was this chapter about?
- Insights, interesting points, new ideas
- What was unclear?
- Any strange words you couldn't understand?

Activity Two: What is Research?

Requirements & Characteristics

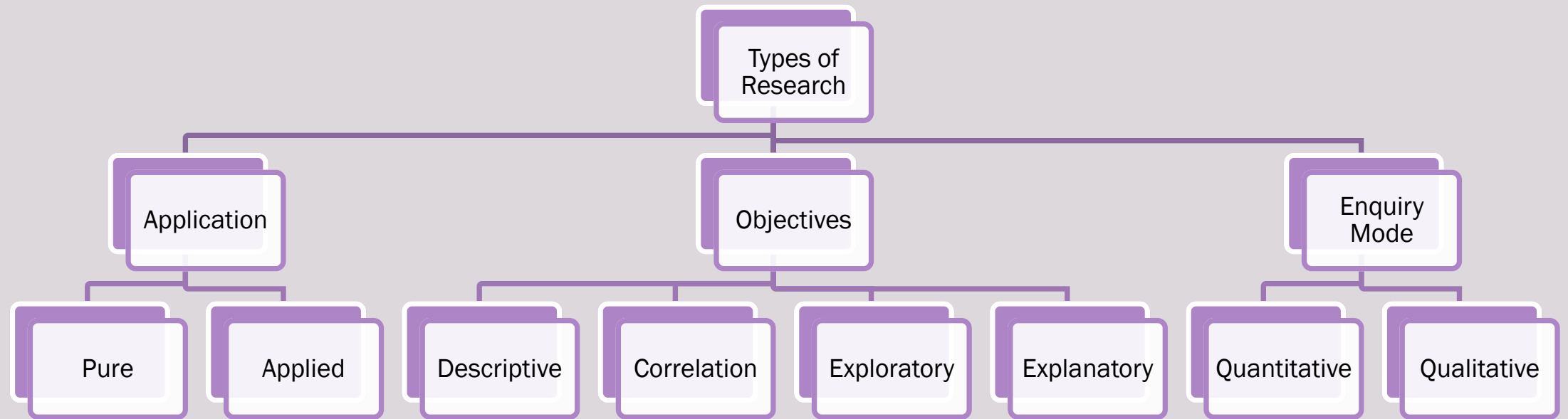
A **process** for collecting, analysing and interpreting information to answer questions.
This process has to be...

1. *Controlled*
 2. *Rigorous*
 3. *Systematic*
 4. *Empirical*
 5. *Valid & verifiable*
 6. *Critical scrutiny*
- In pairs, share your reading notes and write up a description for what each of these characteristics mean

A **process** for collecting, analysing and interpreting information to answer questions. This process has to be...

1. Controlled
Impact of variables on outcomes or quantify its impact
2. Rigorous
Scrupulous adherence to procedures
3. Systematic
Logical sequence, not haphazard
4. Valid & verifiable
Answers are correct, can be verified by you or others (data, quality, plausibility of data)
5. Critical scrutiny
Procedures employed (are the fool-proof, free from drawbacks)
6. Empirical
Collection of hard evidence

Activity Three: Types of Research



- Work with a partner
- Using your reading notes/textbook chapter draft detailed descriptions of each of these aspects/descriptors – also see Table 1.1
- Provide examples for each descriptor

NEXT WEEK

- Think of a topic/problem you would like to research
 - *What kind of question(s) would you ask?*
 - *Why are you interested in this topic? What are your reasons for undertaking this research (list as many reasons as you can)*
 - *How would you describe the kind of research (types of research) you would undertake to answer the question(s) in a valid way*
 - *BRING your notes to class next week, ready for sharing (LINK to research proposal assignment due in May)*
- Blaxter et al (Chapter 1 & Kumar Chapter 2)
 - *Read so that you can answer the following questions*
 - What are the two chapters about?
 - What are some of the common feelings (concerns or worries) Blaxter et al associated with undertaking a new research project?
 - What is the research process? What does it describe?
 - Look at the different representations of the research process on pages 8&9 of Blaxter et al. List three different ways the research process can be depicted?
 - How do these representations compare with Figure 2.3 in Kumar (2011)?
 - Identify the phases of the research process described by Kumar. Explain the activities undertaken in each phase.